DTU Space
Department of Space Research
and Space Technology

Report on Probabilistic Inversion and Cross Hole GPR Tomography

Jeppe Klitgaard <s250250@dtu.dk>

Abstract

Cross-Hole Ground Penetrating Radar (GPR) tomography presents a highly non-linear and ill-posed inverse problem. We demonstrate the use of the optionally GPU-accelerated probabilistic inversion framework `nanopinv` to efficiently reconstruct 2D spatial distributions of phase velocities from travel-time measurements at the Boise Hydrogeophysical Research Site (BHRS). The forward problem is governed by the eikonal equation, which is solved using a second-order Fast Sweeping Method (FSM). We combine this with a Gaussian spatial prior and a Preconditioned Crank-Nicolson proposal to sample the posterior distribution via the Extended Metropolis algorithm and a Deterministic Even-Odd (DEO) Non-Reversible Parallel Tempering scheme. The method successfully reconstructs the velocity and porosity fields at the site, estimating the Total Water In Place (TWIP) beneath the water table to be $(27.02 \pm 0.11)\,\mathrm{m}^2$.

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Artificial Intelligence Declaration

The body of work presented below is solely authored by the author of this document, Jeppe Klitgaard, and none of the presented material has been produced by means of *Generative AI* in the manner requiring citation as described by [1] outside the entries listed below.

Generative AI has, however, been used selectively in the generation of limited parts of the code. GitHub Copilot tab-completion has been used during the development of the attached code and in the generation of an initial version of the following entries:

- Some parts of `nanopinv` (which falls outside the examined material of this report), in particular:
  - A number of implementations of the Fast Sweeping Method
  - Boiler-plate code around benchmarking and testing these implementations
  - Initial implementations for the plotting methods supported by the `History` class

The generated code output has in all instances been checked and often revised by the authors and is *not* considered to be a *significant* contribution to total intellectual work undertaken and presented.

1 Introduction

This report seeks to demonstrate the use of probabilistic inversion techniques to solve the non-linear inverse problem of Ground Penetrating Radar (GPR) Tomography in which the spatial distribution of phase velocities through the subsurface are reconstructed from observed travel times of radar waves between sources and receivers located in two nearby boreholes.

In order to produce the results presented in this report, we have developed a performant Python library, `nanopinv`, which is able to achieve accurate reconstructions of the modelled field by sampling the posterior distribution of the modelled field using Markov Chain Monte Carlo (MCMC) methods. The library is implemented in JAX [2] and can efficiently leverage multiple CPU cores or GPU hardware.

The performance of the library is further enhanced by implementations of the Fast Sweeping Method (FSM) [3] for solving the eikonal equation and enables users to rapidly obtain independent samples by leveraging the Parallel Tempering method.

The structure of the report deviates from the typical layout of an experimental report and instead interleaves theory and results. This choice is made in the interest of sparing the reader from the rather hefty theory section featured in the first draft of the report. Further, the exploratory and interdisciplinary nature of the project incentivises a less rigid structure in which theory and design choices are introduced as they become relevant to the discussion of the results. Alternatively, the reader may simply regard Section 3 as a lengthy theory section with accompanying examples, which is followed by more focused results and discussions sections in Section 4 and Section 5 respectively.

2 Data and Experimental Design

The dataset used in this report consists of $N_d = 758$ travel time measurements taken using pairs of $N_{\text{source}} = 52$ source locations and $N_{\text{receiver}} = 235$ receiver locations in two boreholes at the Boise Hydrogeophysical Research Site (BHRS) near Boise, Idaho in the United States. The boreholes are separated by approximately 10 metres and measurements are obtained for elevations in the range $y \in [831.51\,\text{m}, 846.79\,\text{m}]$ as shown on Figure 1.

During the experiment measurements were obtained using the technique of Cross Hole Ground Penetrating Radar (GPR), where the delays in the observed travel times of radar waves emitted by the source and received by its corresponding network of receivers were recorded for each source-receiver pair.

Additionally, we are given 3 samples of subsurface velocity fields shown in Figure 3, which serve as a source of prior information about the spatial distribution of velocities. Lastly, we are informed that the water table at the site is known to be located at an elevation of $y_* = 844\,\text{m}$.

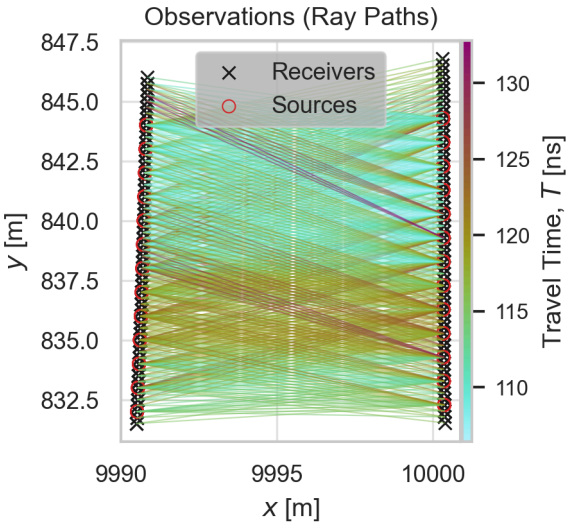


Figure 1: Source and receiver locations with observed travel times overlaid on the model space.

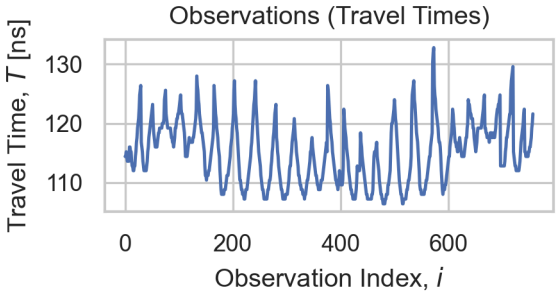


Figure 2: Observed travel times for all source-receiver pairs.

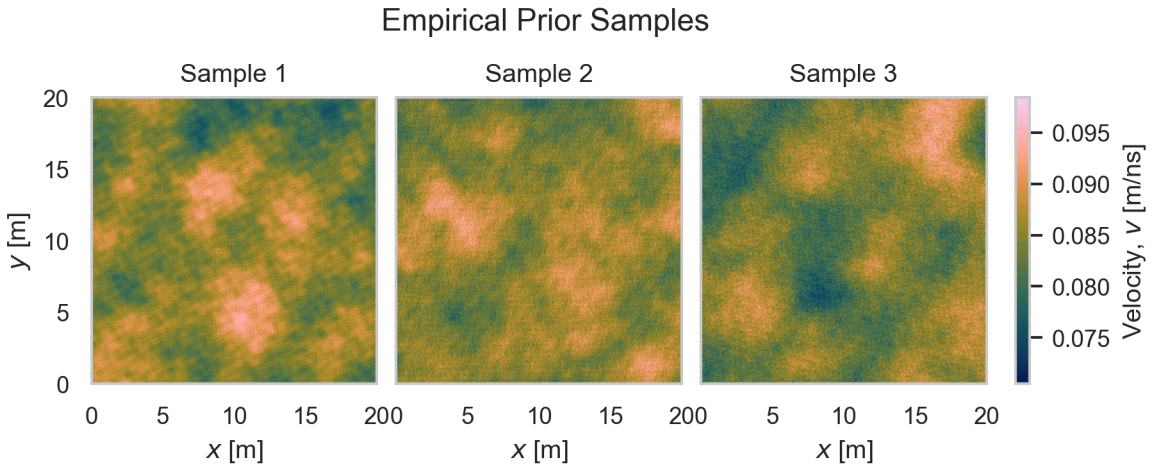


Figure 3: The three samples representing the prior knowledge of the velocity field.

3 Theory and Exploration

In order to arrive at a suitable theoretical framing for the treatment of the inverse problem in question, we introduce the relevant concepts as the data is explored while validating and motivating the theory through presentation and brief discussion of intermediate results.

3.1 Probabilistic Inversion

We start by introducing the larger structures within the framework of *probabilistic inversion*, to which end we adopt the notation due to Tarantola [4]. This choice of framework is convenient for travel-time tomography problems as it handles non-linearities in the forward model naturally, while also offering a principled way to incorporate prior information and uncertainty quantification into the inversion.

We may tersely summarise the framework as presented in [5, 4.2] by letting $\rho_m(\boldsymbol{m}) : \mathcal{M} \to \mathbb{R}$ and $\rho_d(\boldsymbol{d}) : \mathcal{D} \to \mathbb{R}$ denote the *prior model probability distribution* and *prior data probability distribution* respectively, with their arguments denoting the model parameters and data. For the problem covered by this report, the model space is 2-dimensional, $\mathcal{M} = \mathbb{R}^{N_x \times N_y}$, while the data space is 1-dimensional, $\mathcal{D} = \mathbb{R}^{N_d}$. By assuming the data to be independent of the model parameters, we construct a joint prior distribution:

$$\rho(\boldsymbol{d}, \boldsymbol{m}) = \rho_d(\boldsymbol{d})\rho_m(\boldsymbol{m}). \tag{1}$$

The forward model is then introduced through $\boldsymbol{d} = g(\boldsymbol{m}) + \boldsymbol{e}$ where the error $\boldsymbol{e}$ captures any numerical errors in the computation of the forward model $g(\boldsymbol{m})$. Additionally, any potential stochasticity in the forward model and systematic biases such as those introduced by the simplifying assumptions made in the physical model as discussed in Section 3.3 enter through this error term. We may capture these effects by another joint probability distribution, $\theta(\boldsymbol{d}, \boldsymbol{m})$, that links the model and data spaces through the forward model.

We introduce the notion of *homogenous probability density functions*, $\mu(\cdot)$, which have probability densities proportionate to the volume element of the space on which they are defined. As such, they may be used to ensure correct normalization of probability distributions under coordinate transformations and notably do not carry any information on their own.

With this in mind, we relate the joint linking distribution $\theta(\boldsymbol{d}, \boldsymbol{m})$ to its conditional through the product rule to obtain

$$\begin{aligned} \theta(\boldsymbol{d}, \boldsymbol{m}) &= \theta(\boldsymbol{d}|\boldsymbol{m})\theta(\boldsymbol{m}) \\ &= \theta(\boldsymbol{d}|\boldsymbol{m})\mu_m(\boldsymbol{m}) \end{aligned} \tag{2}$$

where $\theta(\boldsymbol{m})$ is identified with $\mu_m(\boldsymbol{m})$ on the grounds that neither carry any information about the model parameters. This can be understood as $\theta(\boldsymbol{d}, \boldsymbol{m})$ not describing the model parameters themselves, only how the data $\boldsymbol{d}$ is linked to them.

As shown in [4, Eq. 1.83], these distributions may be combined to obtain the *joint posterior model distribution*:

$$\sigma(\boldsymbol{m}, \boldsymbol{d}) = k\frac{\rho(\boldsymbol{m}, \boldsymbol{d})\theta(\boldsymbol{d}, \boldsymbol{m})}{\mu(\boldsymbol{m}, \boldsymbol{d})}, \tag{3}$$

where k is a normalisation constant and $\mu(\boldsymbol{m}, \boldsymbol{d})$ is the *joint homogenous probability density function*. This further simplifies to the *posterior model distribution* under the following treatment

$$\begin{aligned}
\sigma_m(\boldsymbol{m}) &= k\int_{\mathcal{D}} \frac{\rho(\boldsymbol{m}, \boldsymbol{d})\theta(\boldsymbol{d}, \boldsymbol{m})}{\mu(\boldsymbol{m}, \boldsymbol{d})}\,\mathrm{d}\boldsymbol{d} && \text{Margin. over data space, } \mathcal{D} \\
&= k\rho_m(\boldsymbol{m})\int_{\mathcal{D}} \frac{\rho_d(\boldsymbol{d})\theta(\boldsymbol{d}, \boldsymbol{m})}{\mu(\boldsymbol{m}, \boldsymbol{d})}\,\mathrm{d}\boldsymbol{d} && \text{Equation 1} \\
&= k\rho_m(\boldsymbol{m})\int_{\mathcal{D}} \frac{\rho_d(\boldsymbol{d})\theta(\boldsymbol{d}, \boldsymbol{m})}{\mu_m(\boldsymbol{m})\mu_d(\boldsymbol{d})}\,\mathrm{d}\boldsymbol{d} && \text{Assume } \mu(\boldsymbol{m}), \mu(\boldsymbol{d}) \text{ independent} \\
&= k\rho_m(\boldsymbol{m})\int_{\mathcal{D}} \frac{\rho_d(\boldsymbol{d})\theta(\boldsymbol{d}|\boldsymbol{m})\cancel{\mu_m(\boldsymbol{m})}}{\cancel{\mu_m(\boldsymbol{m})}\mu_d(\boldsymbol{d})}\,\mathrm{d}\boldsymbol{d} && \text{Equation 2} \\
&= kL(\boldsymbol{m})\rho_m(\boldsymbol{m}) && L(\boldsymbol{m}) := \int_{\mathcal{D}} \frac{\rho_d(\boldsymbol{d})\theta(\boldsymbol{d}|\boldsymbol{m})}{\mu_d(\boldsymbol{d})}\,\mathrm{d}\boldsymbol{d}
\end{aligned} \tag{4}$$

where $L(\boldsymbol{m})$ is the *likelihood probability density* describing how likely the observed data $\boldsymbol{d}$ is given the model parameters $\boldsymbol{m}$.

3.1.a Likelihood

If we assume the data to be independently and normally distributed and additionally assume the same of the errors introduced by the forward model, both $\rho_d(\boldsymbol{d})$ and $\theta(\boldsymbol{d}|\boldsymbol{m})$ can be expressed as independent normal distributions:

$$\begin{aligned}
\rho_d(\boldsymbol{d}) &= \mathcal{N}(\boldsymbol{d}|\boldsymbol{\mu}_d, \Sigma_d) \\
\theta(\boldsymbol{d}|\boldsymbol{m}) &= \mathcal{N}(\boldsymbol{d}|g(\boldsymbol{m}), \Sigma_e)
\end{aligned} \tag{5}$$

which convolve to give the likelihood as another normal distribution:

$$L(\boldsymbol{m}) \propto \mathcal{N}(g(\boldsymbol{m})|\boldsymbol{\mu}_d, \Sigma_d + \Sigma_e). \tag{6}$$

We further assume the errors of the forward model to be negligible compared to the variance of the data distribution, which is known a priori to be 4 ns. That is, given $\Sigma_e \ll \Sigma_d$, the likelihood can be approximated as

$$L(\boldsymbol{m}) \propto \mathcal{N}(g(\boldsymbol{m})|\boldsymbol{\mu}_d, \Sigma_d). \tag{7}$$

3.2 Prior Information and 2-Point Statistics

In order to turn the given velocity field samples shown in Figure 3 into a usable prior distribution, realise that 1-point statistics will not suffice due to the strong spatial correlation in the field. Instead, we turn to 2-point statistics wherein the covariance between pairs of points in the field can be modelled. By assuming that the covariance of the field is isotropic and the process it is drawn from to be stationary, we propose that the covariance between any two points in the field depends only their separation, h. This leads to the notion of *covariance functions* of the form $C(h)$, with a notable example being the *spherical covariance function*:

$$C_{\text{sph}}(h) = \begin{cases} \sigma^2\left(1 - \frac{3h}{2s} + \frac{h^3}{2s^3}\right) & h \leq s \\ 0 & h > s \end{cases} \tag{8}$$

where the *variance*, σ^2, and *range*, s, are hyperparameters of the covariance function.

In order to assess the suitability of a covariance function for a given set of samples, we may estimate the empirical *semi-variogram*, $\hat{\gamma}(h)$ of the samples by binning the separations into N bins of width Δh:

$$\hat{\gamma}(h_k \pm \Delta h) = \frac{1}{2N_k} \sum_{(i,j) \in S_k} \left(\boldsymbol{v}_i - \boldsymbol{v}_j\right)^2 \tag{9}$$

where h_k is the bin centre, $S_k = \left\{(i,j) : h_k - \Delta h < \left\|\boldsymbol{v}_i - \boldsymbol{v}_j\right\| < h_k + \Delta h\right\}$ is the set of indices over the point pairs in the bin and N_k is the number of elements in S_k. The empirical semi-variogram can then be obtained by computing $\hat{\gamma}(h_k \pm \Delta h)$ for each bin and averaging across the samples.

Such a semi-variogram is related to the covariance function by

$$\gamma(h) = \frac{1}{2}(\sigma^2 - C(h)) \tag{10}$$

where notably $\gamma(h)$ here represents an analytical, exact semi-variogram.

3.2.a Obtaining a prior distribution

In order to obtain a covariance function that suitably captures the spatial correlations in the prior samples, we may perform a fit of candidate covariance functions $C(h;\theta)$ in which their hyperparameters θ are determined by minimising the mean squared error between the empirical semi-variogram and the analytical semi-variogram implied by the covariance function:

$$\theta^* = \arg\min_{\theta} \left\| \hat{\gamma}(h) - \frac{1}{2}(\sigma^2 - C(h;\theta)) \right\|_2^2 \tag{11}$$

By first computing the empirical semi-variogram of the prior samples from Figure 3 and subsequently fitting a variety of candidate covariance functions to it, we are able to pick a suitable function. The empirical semi-variogram is estimated using `gstools` [6], which is also used to perform the fitting. Notably, the global tolerance parameter of the underlying fitting routine is decreased to 10^{-14} as the fits were otherwise observed to be poor.

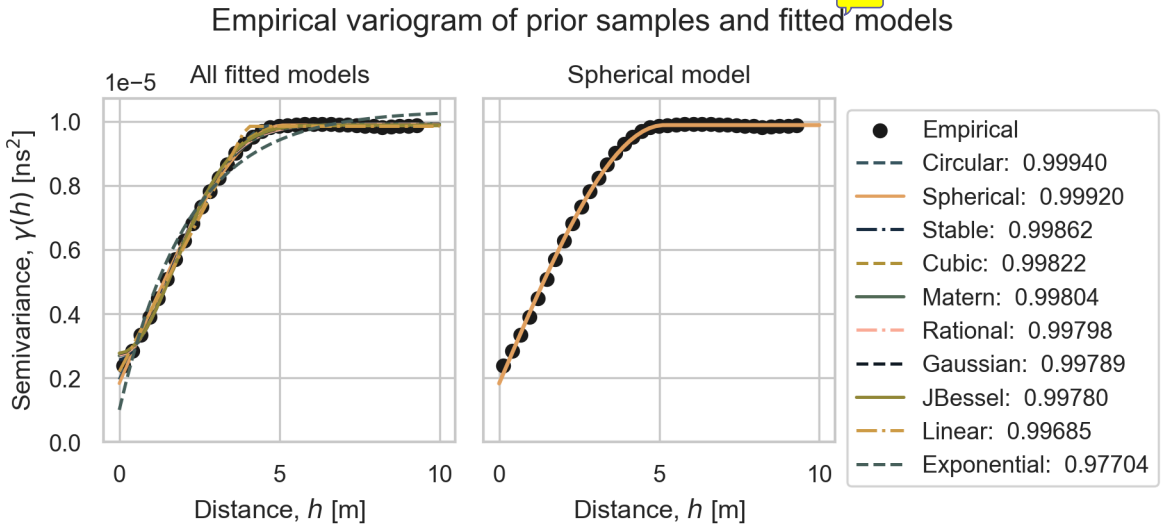


Figure 4: Empirical semi-variogram of the prior samples and fits of candidate covariance functions.

While the circular covariance function slightly outperforms the spherical covariance function in terms of the R^2 metric, we select the spherical covariance function for our prior distribution as it appears to be a more standard choice in the literature. The parameters of the fitted spherical covariance function are found to be $\sigma^2 = 8.06 \times 10^{-6}$, $s = 5.16$, $\tau^2 = 1.83 \times 10^{-6}$ where τ^2 is the *nugget*, which represents the variance at zero separation and is simply added to the covariance function as $C(h) + \tau^2$.

In order to obtain a prior distribution using two-point statistics, we also need to obtain an estimate of the mean of the distribution. As we have already assumed isotropy, we let the mean be constant across the field and estimate it by averaging across the prior samples to obtain $\mu = 0.084\,9$.

With estimates of both the mean and the covariance function, we are able to construct a Gaussian prior distribution over the model space. In practice, this involves constructing the full covariance matrix, which for a model discretisation of $\Delta x = \Delta y = 0.25\,\text{m}$ gives a model grid of size $N_x = 41$, $N_y = 63$ and covariance of size 2583×2583. The efficiency of the prior sampling in `nanopinv` is enhanced by performing a Cholesky decomposition of the covariance matrix to obtain the lower triangular matrix $\mathbf{L}$ during initialisation. Samples are then produced by sampling a standard normal distribution $z \sim \mathcal{N}(0, 1)$ and applying the affine transformation $x = \mu + Lz$.

To gauge whether the obtained prior distribution is a plausible representation of the prior samples, we draw 3 samples from the distribution and compare them to the original samples in Figure 5. The drawn samples appear to be similar in the characteristic length scale and magnitude of the variations, though notably does not reproduce the small-scale, directional patterns that appear to be present in the original samples. These violate the assumption of isotropy and may further be speculated to be systematic errors from the experiment.

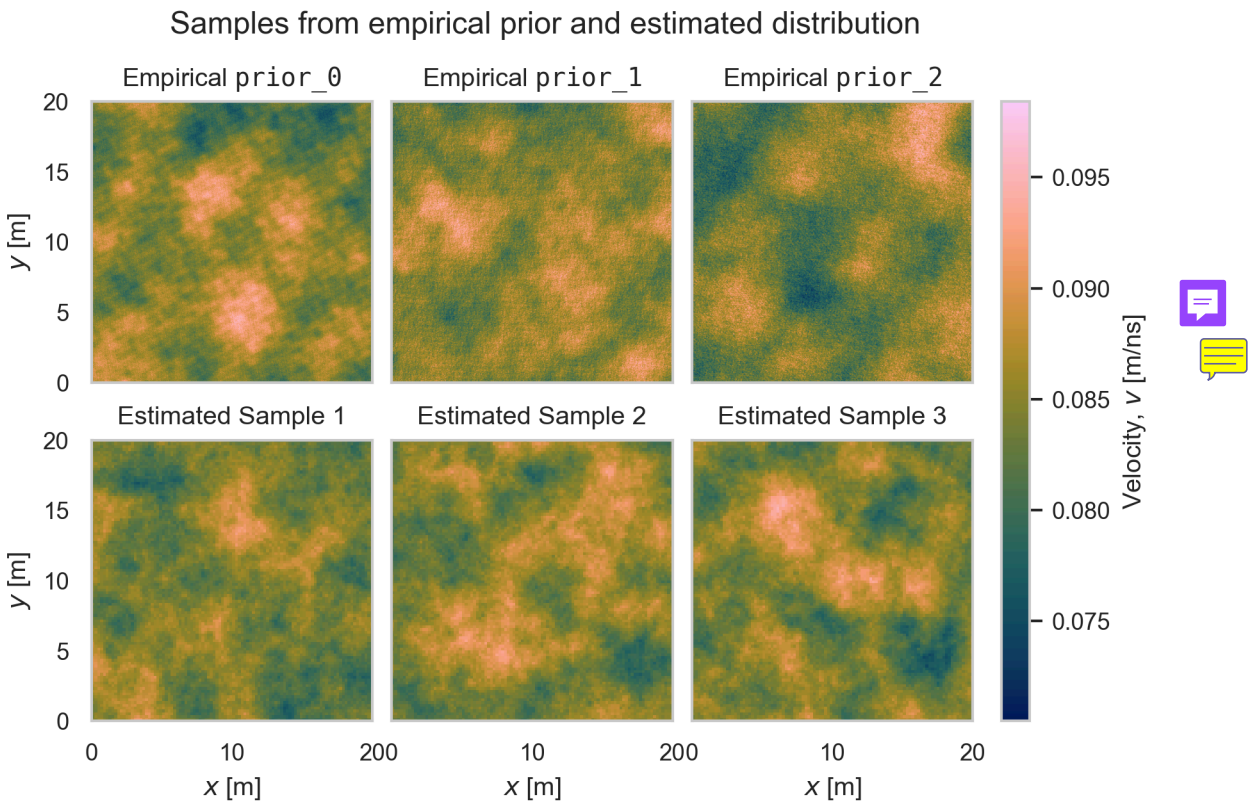


Figure 5: Comparison of samples from the obtained Gaussian prior distribution and those of the empirical prior.

3.3 Physics and Forward Model

Having produced a prior distribution over the model space, we turn to the physics involved in the ground penetrating radar experiment in order to obtain a suitable forward model, $g : \mathcal{M} \to \mathcal{D}$, that maps the model parameters to the space of observations.

3.3.a Eikonal Equation and Wave Travel Time

In order to relate the observed travel times to the spatial distribution of the phase velocity, we refer to [7, 3.1.1] in which the *eikonal equation*, Equation 12, is derived from Maxwell's equations under a number of assumptions.

$$\|\nabla S(\boldsymbol{r})\|_2^2 = n^2(\boldsymbol{r}) \tag{12}$$

where $S : \mathbb{R}^N \to \mathbb{R}$ is the *optical path* as a function of position $\boldsymbol{r}$ and n is the *refractive index*. We may relate this to the travel time $T(\boldsymbol{r})$ and phase velocity $v(\boldsymbol{r})$ by noting that the optical path is defined as the distance travelled in vacuum, $S(\boldsymbol{r}) = c_0 T(\boldsymbol{r})$ and recalling the definition of the refractive index, $n(\boldsymbol{r}) = \frac{c_0}{v(\boldsymbol{r})}$, which yields

$$\|\nabla T(\boldsymbol{r})\|_2 = \frac{1}{v(\boldsymbol{r})}. \tag{13}$$

The derivation due to Born and Wolf relies on the assumption that the characteristic length scale of the variations in the refractive index — or equivalently, the phase velocities — are much larger than the wavelengths of the electromagnetic waves. If we assume the data to be collected with similar equipment to that used as part of the study described in [8], we may obtain estimates of the wavelengths of the radar waves with which we can reason about the validity of the high-frequency assumption, $\lambda \ll L$, where L is the characteristic length scale of variations in the modelled field through the medium.

The study cites an antenna frequency of $f = 100\,\text{MHz}$, with relative permittivities $\varepsilon/\varepsilon_0$ of $\approx \{1, 4, 81\}$ [F/m] for air, soil, and water respectively. We may find the wavelengths associated with these parameters using the relation

$$\lambda = \frac{v}{f} = \frac{c_0}{f\sqrt{\varepsilon/\varepsilon_0}}, \tag{14}$$

which gives $\lambda_{\text{soil}} = 1.5\,\text{m}$ and $\lambda_{\text{water}} = 0.3\,\text{m}$. These effectively set a lower bound on the size of the features that can be resolved using the acquired data. This is an valuable metric to keep in mind, as it also provides a useful scale of discretisation for the model space. In particular, no additional resolution can be obtained by further refining the model discretisation beyond this limit without also increasing the frequency of the radar waves or resorting to a more complex physical model that relaxes the high-frequency assumption. As such, the previously proposed discretisation of $\Delta x = \Delta y = 0.25\,\text{m}$ may already be below the limit of resolution set by the GPR frequency and the properties of the medium.

3.3.b Numerical Solution of the Forward Problem

In order to compute the forward projection of a given model, here understood to be a discretised spatial distribution of phase velocities, we must solve Equation 13, and ideally do so as efficiently as possible.

A popular choice of algorithm dedicated to solving the family of equations to which the eikonal equation belongs is the Fast Marching Method (FMM) due to Sethian [9], which is implemented as first and second-order methods in [10]. The finer details of the FMM algorithm is beyond the scope of this report, but a notable consequence of the formulation of the algorithm is that it is not amenable to parallelisation. While multiple source-receiver pairs may be computed in parallel using CPU cores, the method is not able utilise modern, massively-parallel hardware such as GPUs efficiently.

For this reason, the alternative Fast Sweeping Method (FSM) due to Zhao [3], [11] is considered a superior choice for solving the eikonal equation on GPUs, as it more naturally lends itself to

parallelisation. Taking the paper [12] as a reference, `nanopinv` implements both a first and second-order version of the parallel FSM algorithms suitable for use with CPU or GPU hardware.

Common to all the implementations is that they produce an approximation of the travel time T from a source $\boldsymbol{r}_s$ to all other points in the model space when given a propagation velocity field $v(\boldsymbol{r})$. The estimated travel time to the receivers, $\boldsymbol{r}_r$, may then be obtained by interpolation. Figure 6 shows the forward projections obtained by the 2nd order FSM solver for three samples drawn from the prior distribution obtained in Section 3.2.a.

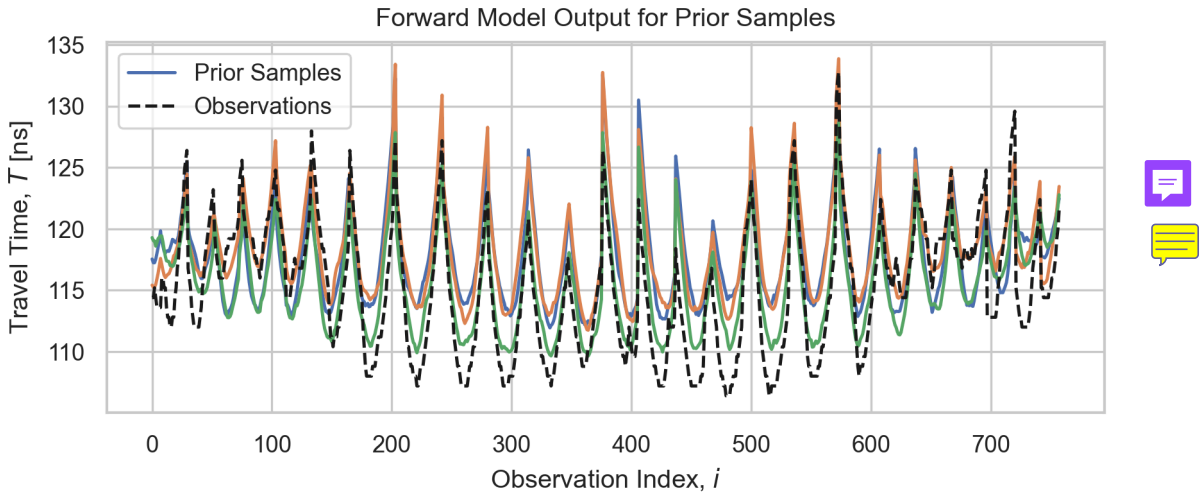


Figure 6: Comparison of the travel time fields obtained by the first and second-order FSM implementations for a given velocity field.

3.4 Markov Chain Monte Carlo Methods

In order to sample the posterior, $\sigma_m(\boldsymbol{m})$, we turn to the theory of Markov Chains and Monte Carlo sampling, in which a chain of samples is generated by a Markov process — that is, a process in which the next state depends only on the current state.

While many algorithms for simulating such processes exist, we will focus on the *Extended Metropolis algorithm* due to Mosegaard and Tarantola [13] whose transition procedure is described in Algorithm 2. In order to efficiently sample the prior model distribution even for large model spaces, we leverage the Preconditioned Crank-Nicolson proposal algorithm given in Algorithm 1.

It should be noted that implementations of Algorithm 2 typically work with the log-transformed likelihood and ensure the forward model and likelihood are only evaluated once per set of model parameters by to improve performance.

Algorithm 1: Preconditioned Crank-Nicolson Proposal

procedure PRECONDITIONEDCRANKNICOLSONPROPOSAL($\boldsymbol{m}_0$, δ, ρ_m)
  input Initial model $\boldsymbol{m}_0 \in \mathcal{M}$,
    Step size $\delta \in (0, 1)$,
    Prior distribution $\rho_m : \mathcal{M} \to \mathbb{R}$
  output Proposed model $\boldsymbol{m}_p \in \mathcal{M}$
  $\boldsymbol{\mu} \leftarrow \mathbb{E}[\rho_m]$ ▷ Mean of prior model distribution
  $\boldsymbol{m}_* \leftarrow \boldsymbol{x} \sim \rho_m$ ▷ Sample from prior model distribution
  $\boldsymbol{m}_p \leftarrow \boldsymbol{\mu} + \sqrt{1-\delta^2}(\boldsymbol{m}_0 - \boldsymbol{\mu}) + \delta(\boldsymbol{m}_* - \boldsymbol{\mu})$ ▷ Preconditioned Crank-Nicolson
  return $\boldsymbol{m}_p$
end

Algorithm 2: Extended Metropolis Algorithm

procedure ExtendedMetropolisStep($\boldsymbol{m}_0$, β, q, g, L)
  input Initial model $\boldsymbol{m}_0 \in \mathcal{M}$, Inverse temperature $\beta \in (0, 1]$,
    Proposal algorithm $q : \mathcal{M} \to \mathcal{M}$,
    Forward model $g : \mathcal{M} \to \mathcal{D}$, Likelihood $L : \mathcal{D} \to \mathbb{R}$,
  output Final model $\boldsymbol{m} \in \mathcal{M}$, Acceptance $\in \{\text{TRUE}, \text{FALSE}\}$
  $\boldsymbol{m}_p \leftarrow q(\boldsymbol{m}_0)$ ▷ Propose new model
  $P_{\text{accept}} \leftarrow \min\left(1, \left(\frac{L(g(\boldsymbol{m}_p))}{L(g(\boldsymbol{m}_0))}\right)^{\beta}\right)$ ▷ Compute acceptance probability
  $\alpha \leftarrow x \sim \mathcal{U}(0, 1)$ ▷ Sample uniform distribution
  if $\alpha < P_{\text{accept}}$ **then**
    return $\boldsymbol{m}_p$, TRUE ▷ Accept proposal
  else
    return $\boldsymbol{m}_0$, FALSE ▷ Reject proposal
  end
end

3.4.a Sampling the Posterior

By using these algorithms with sufficient care, we will be able to sample from the posterior distribution of the model parameters, $\sigma_m(\boldsymbol{m})$. Firstly, the choice of the step size, δ, will have significant influence on the efficiency with which we can sample. By inspecting the proposal algorithm in Algorithm 1, it is clear that a larger step size will retain less of the current model and instead be more heavily influenced by the sample drawn from the prior distribution. As such, a larger step size will more aggressively explore the model space. This is of course desirable, but comes at the cost of a lower acceptance rate as many of the proposed models will inevitably lie in regions of low likelihood inside the model space. Conversely, if the chosen step size is too small, the acceptance rate will be high, but the chain will be slow to explore the model space and will be more likely to get stuck in local valleys within the model space, thus failing to accurately sample the posterior. The literature generally favours a step size corresponding to an acceptance rate of approximately 25%.

The second important hyperparameter is the inverse temperature, $\beta \in (0, 1]$, which directly affects the acceptance probability of the proposed models. As β goes towards zero, the acceptance probability approaches unity, and as such the inverse temperature may be understood to effectively flatten the likelihood landscape thus making it easier to traverse. This comes with the unfortunate consequence that samples drawn during simulations away from unit inverse temperature are not distributed according to the true posterior distrubution. As such, we concern ourselves only with *cold sampling* at $\beta = 1$ for now.

Further, we must only consider *accepted* samples as being drawn from the posterior distribution, and must additionally take care that the drawn samples are sufficiently decorrelated from one another as they would otherwise bias our estimate of the posterior distribution.

We set up the `ExtendedMetropolisChain` sampler from `nanopinv` with the proposal algorithm given in Algorithm 1 using the previously obtained prior distribution $\rho_m(\boldsymbol{m})$ and a step size of $\delta = 0.05$. An initial model $\boldsymbol{m}_0$ is drawn directly from the prior distribution and the $N_{\text{iter}} = 1\,000$ steps are taken while recording intermediate samples and acceptance decisions. It is highly likely that the initial model drawn from the prior distribution will lie in a region of low likelihood, which leads to the burn-in phenomenon shown in Figure 7. Here the log-likelihood is observed to decrease rapidly during the initial steps of the chain before stabilising as the chain diffuses towards a higher likelihood region of the model space.

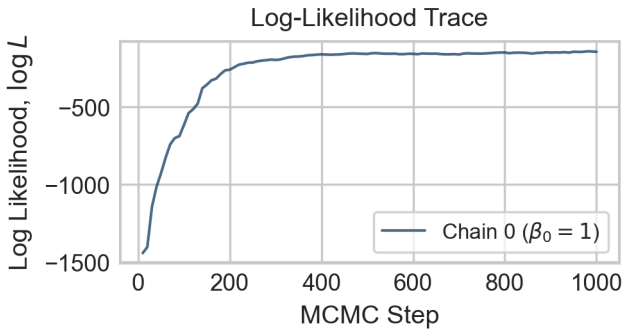


Figure 7: Trace log likelihood during the burn-in phase of the Extended Metropolis algorithm.

Using JAX it is particularly easy to run multiple chains in parallel using the `vmap` function. This is useful as it allows us to obtain multiple independent samples, and additionally is a useful way to gauge whether the chains get stuck in local minima by comparing the results of multiple chains initialised with different random seeds. By performing $N_{\text{burn-in}} = 3\,000$ steps on $N_{\text{chains}} = 10$ chains, and subsequently recording another $N_{\text{samples}} = 10\,000$ samples, we are able to produce a large number of samples. Running the cumulative 100 000 steps takes approximately 6 minutes and 30 seconds on an NVIDIA RTX 4090 GPU.

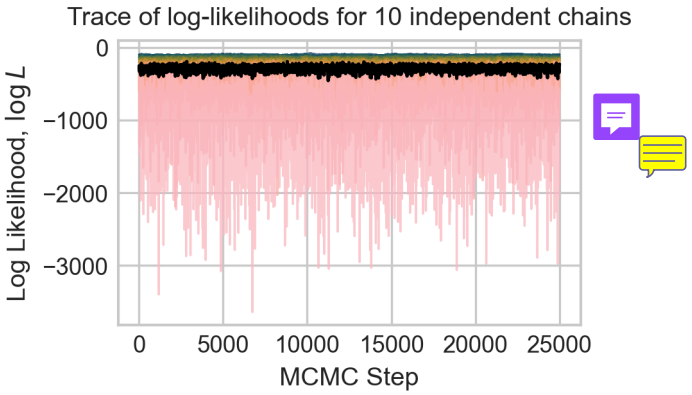


Figure 8: Trace of log likelihood values for 10 chains during sampling after burn-in.

Inspecting Figure 8 reveals that all chains appear to have converged to a similar level set of the likelihood landscape, which is an indication that the chains are not getting stuck in local minima. Figure 9 shows that the acceptance rate is observed to be around 45% during the sampling, which is higher than the 25% target. This can negatively affect the correlation times and heightens the risk of poor mixing.

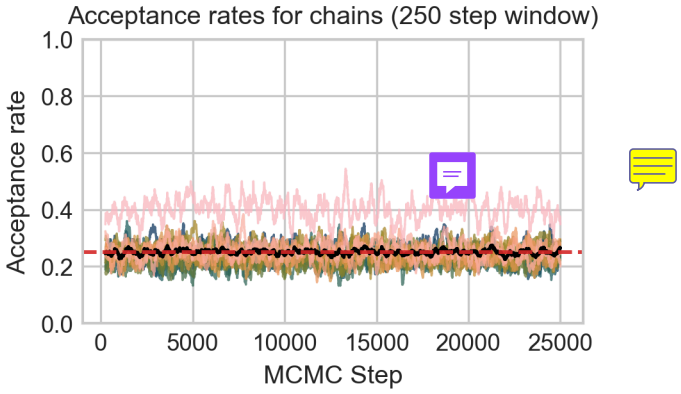


Figure 9: Trace of acceptance rate for samples during sampling. Red dashed line indicates 25% and black line denotes mean acceptance rate across all chains.

As observed in Figure 10, the autocorrelation of the likelihood roughly follows a decaying exponential with a characteristic decay length of roughly 700 steps. We note with caution that the autocorrelation of the *likelihood* is not perfectly representative of the autocorrelation of the *model parameters* and may be artificially longer in cases where the proposal along contours of the likelihood landscape. In order to obtain uncorrelated samples we must thin the accepted samples such that the number of steps between recorded samples is greater than some small multiple of the decay length.

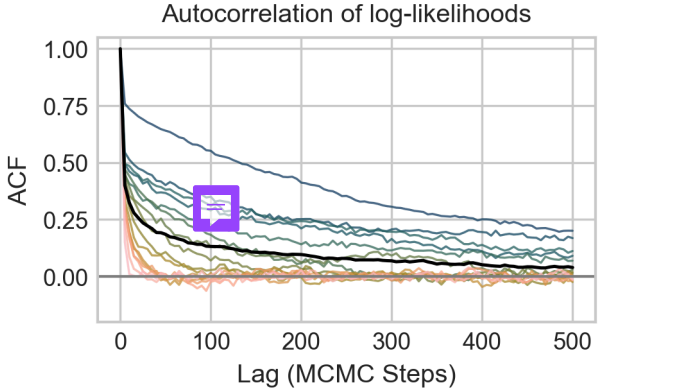


Figure 10: Autocorrelation of the samples during sampling. Black line denotes mean across all chains.

By requiring a minimum interval of 2 000 steps between recorded samples we obtain a total of 50 uncorrelated samples from the posterior distribution, 3 of which are shown in Figure 11.

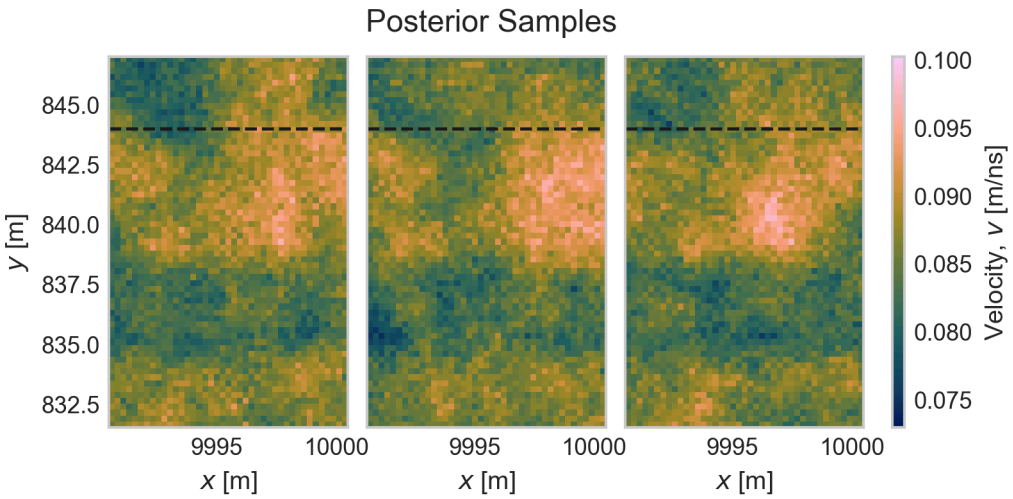


Figure 11: Posterior samples obtained by the Extended Metropolis algorithm.

3.4.b Step size tuning

As mentioned previously, the choice of step size has significant implications for the efficiency of the sampling. For this reason, `nanopinv` additionally implements a simple step size tuning procedure in which the step size is adjusted iteratively by using batches of N_{batch} samples to compute the update according to the Robbins-Monro algorithm:

$$\delta_{k+1} = \delta_k + \xi_k\left(\bar{P}_k - P_{\text{target}}\right) \qquad \text{where} \qquad \xi_k = \frac{\xi_0}{(k+1)^{-\kappa}} \tag{15}$$

where $\xi_k \in (0, 1]$ is a *learning rate* which we decay from a base value ξ_0 according to the batch number k and a decay parameter $\kappa \in (1/2, 1]$. P_{target} denotes the target acceptance rate and $\bar{P}_k$ is the observed acceptance rate for the kth batch of samples.

This is implemented in `nanopinv` via the `tune` method of the `ExtendedMetropolisChain` class, which for $\kappa = 0.5, \xi_0 = 0.1, N_{\text{batch}} = 1\,000$ produces the tuning history shown in Figure 12 and converges to a step size of $\delta = 0.808$ with a corresponding acceptance rate of approximately 25%.

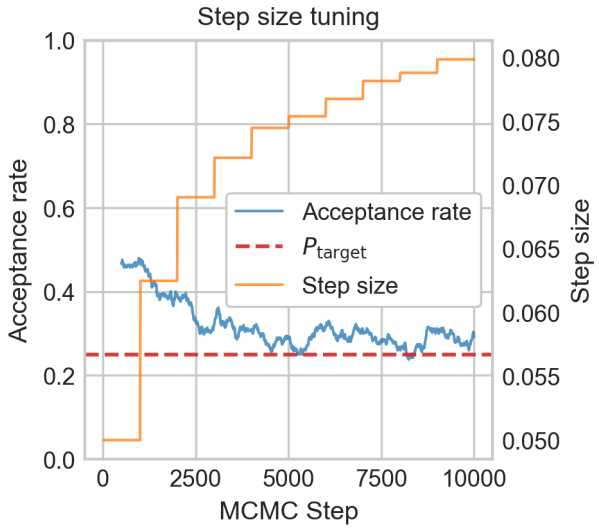


Figure 12: Trace of step size during tuning procedure. Red dashed line indicates the initial step size and black dashed line indicates the final step size after tuning.

3.4.c Parallel Tempering

The primary bottle-neck of our current simulations are the long correlation times between the samples. While running multiple chains in parallel is a useful workaround, the performance can be further improved by implementing a *parallel tempering* scheme. In Section 3.4.a we discussed how the inverse temperature, β, can flatten the likelihood landscape, thus allowing much larger steps to be taken without compromising the acceptance rate. With this in mind, we can arrange a ladder of chains with increasing temperatures and individual step sizes. Hotter chains will be able to easily

traverse any hills in the likelihood landscape and reach all parts of the model space. Unfortunately, the samples drawn from these hotter chains still cannot be used to garner information about the posterior distribution.

As a result, we propose to swap the states of neighbouring chains on the ladder according to a criterion that preserves *detailed balance* such that the chains remain in equilibrium at all times. This allows the hot chains to lead the charge in exploring the model space while the coldest chain is still able to draw samples from the true posterior distribution. While the full theory behind the parallel tempering scheme is beyond the scope of this report, an accessible description by Paweł Czyż can be found in [14].

In particular, the variant of parallel tempering that is implemented in `nanopinv` is the *Deterministic Even Odd* (DEO) *non-reversible parallel tempering* (PT) scheme due to Syed et al. [15], which is outlined in Algorithm 3 where the DEO variant arises when the parity offset Ξ is alternated between 0 and 1 for each swap. This approach has the limitation that chain states can only diffuse by at most one rung on the ladder per swap, which puts a lower bound on the number of swaps required to traverse the ladder and thus the correlation time of the samples.

With this in mind, the authors of [15] introduce the notion of the *communication barrier*. This is used to produce a tuning scheme for the temperature ladder, which is optimally tuned when the swapping rate is uniform across all rungs of the ladder. An implementation of this tuning scheme alongside a Robbins-Monro-based tuning scheme akin to that discussed in Section 3.4.b is implemented in `nanopinv`, closely following that presented in [14].

Algorithm 3: Non-Reversible Parallel Tempering Swap Procedure

procedure AdjacentSwap(N, Ξ, X, L, β)
  input Number of chains N, Parity offset $\Xi \in \{0, 1\}$,
    Chain states $X = (x_0, ..., x_{N-1})$, Likelihoods $L = (L_0, ..., L_{N-1})$,
    Inverse temperatures $\beta = (\beta_0, ..., \beta_{N-1})$
  output Updated states X', Updated likelihoods L', Acceptances $A \in \mathbb{1}^N$
  $A_i \leftarrow$ False $\forall i \in \{0, ..., N-2\}$ ▷ Initialize acceptances
  $X', L' \leftarrow X, L$ ▷ Initialize outputs
  for $i = 0, ..., N-2$ **do**
    if $(i \bmod 2) = \Xi$ **then**
      $\alpha_i \leftarrow \min\left(1, \left(\frac{L_{i+1}}{L_i}\right)^{\beta_i - \beta_{i+1}}\right)$ ▷ Compute acceptance probability
      $u_i \leftarrow u \sim \mathcal{U}(0, 1)$ ▷ Sample uniform distribution
      if $u_i < \alpha_i$ **then**
        $A_i \leftarrow$ True ▷ Accept swap proposal
        $X'_i, X'_{i+1} \leftarrow X'_{i+1}, X'_i$ ▷ Apply state swap
        $L'_i, L'_{i+1} \leftarrow L'_{i+1}, L'_i$ ▷ Apply likelihood swap
      end
    end
  end
  return X', L', A ▷ Return updated chains and acceptance outcomes
end

The parallel tempering scheme combines with Algorithm 1 and Algorithm 2 by first performing a step of the Extended Metropolis algorithm on each chain in parallel and then performing the non-reversible parallel tempering swap procedure on the ladder of chains.

Importantly, a chain state is considered an *acceptable* sample of the posterior distribution if it resides on the *cold* chain after both the step-and-swap *and* has either been locally updated by the proposal algorithm *or* has been swapped with its neighbour. Even if the state has just arrived from a hotter chain, it is still considered an acceptable sample as the swap procedure obeys *detailed balance* and thus the state is still distributed according to the posterior distribution.

While a more elegant approach to jointly tune the step sizes and temperatures of the ladder may exist, we resort to iteratively tuning the step sizes and temperatures separately, which is found to converge well for the data used in this report.

3.4.d Obtaining a Tuned Parallel Tempering Sampler

In order to obtain a tuned parallel tempering sampler we instantiate a ladder of chains and evolve them according to the following procedure:

1. Construct $N_{\text{chains}} = 16$ chains:
   1. Initial inverse temperatures: $\beta_k = \omega^{-k}$ where $\omega = 2.0 \in [1, \infty)$ is a hyperparameter controlling the initial temperature spacing.
      - As a heuristic, ω should be chosen such that the hottest initial temperature is just hot enough that a chain acceptance rate of 25% is reached for a step size of $\delta = 1$.
      - This ensures that the hottest chain is able to explore the model space entirely unhindered and effectively corresponds to drawing proposals directly from the prior distribution at each step.
      - There is no goldilocks acceptance rate for non-reversible parallel tempering — the acceptance rate will generally be larger for more densely packed temperature ladders, which further lowers the correlation time by improving mixing between the chains.
   2. Initial step sizes: linearly spaced over the range $[0.1, 1.0]$.
   3. Initial states: Draw initial states $\boldsymbol{m}_0$ from prior $\rho_m(\boldsymbol{m})$.
2. Burn in: Perform $N_{\text{burn}} = 500$ steps of the PT scheme.
3. Tune step sizes: Perform 3 batches of $N_{\text{batch}} = 300$ steps of the PT scheme and tune the step sizes according to the Robbins-Monro algorithm with $\kappa = 1/2, \xi_0 = 1$ and target acceptance rate of 25%.
4. Tune temperatures: Perform a single batch of $N_{\text{batch}} = 1\,000$ steps of the PT scheme and tune the temperatures according to the method outlined in [15].
5. Tune step sizes: Perform another 5 batches of tuning.
6. Tune temperatures: Perform another 2 batches of tuning.
7. Tune step sizes: Perform another 5 batches of tuning.
8. Tune temperatures: Perform another 2 batches of tuning.

After tuning the sampler is run for another 1 000 steps to ensure the chains are in equilibrium at the final parameters after which further steps can be recorded as potential samples from the posterior distribution.

3.4.e Validating the `nanopinv` Implementation

In order to validate the implementation of these algorithms in `nanopinv`, we first tune a parallel tempering sampler as described above and record a number, here approximately 100, posterior samples. These are then used to produce an empirical variogram which is then fitted to a spherical covariance function and found to have a variance $\sigma^2 = 1.18 \times 10^{-5}$, range $s = 5.72$, and a nugget $\tau^2 = 1.72 \times 10^{-6}$. The mean of the posterior samples is found to be $\mu = 0.086$. These are then used to produce a new distribution similar to how the prior distribution was obtained in Section 3.2.a and samples are drawn from it. If the implementation is correct, these samples should have similar isotropic, spatial characteristics to true posterior. As such, we may pick a sample that appears physically plausible and where the velocity field is distributed such that it is resolvable by the waves

sent between the sources and receivers — due to their positioning in this experiment horizontal features are more easily resolved than vertical ones.

If we then draw such a suitable synthetic sample from the produced distribution and compute its forward model, we obtain a new set of synthetic travel time observations with which, we can again run the full inversion process using the existing prior and likelihood to see how well this synthetic model can be recovered. To sample the posterior in the synthetic experiment, we tune a parallel sampler using a setup similar t[illegible]at desc[illegible]d above — though with shorter tuning and fewer chains. We then run $N_{\text{steps}} = 10\,000$ steps and thin the samples appropriately to produce the samples underpinning the reconstruction shown in Figure 13.

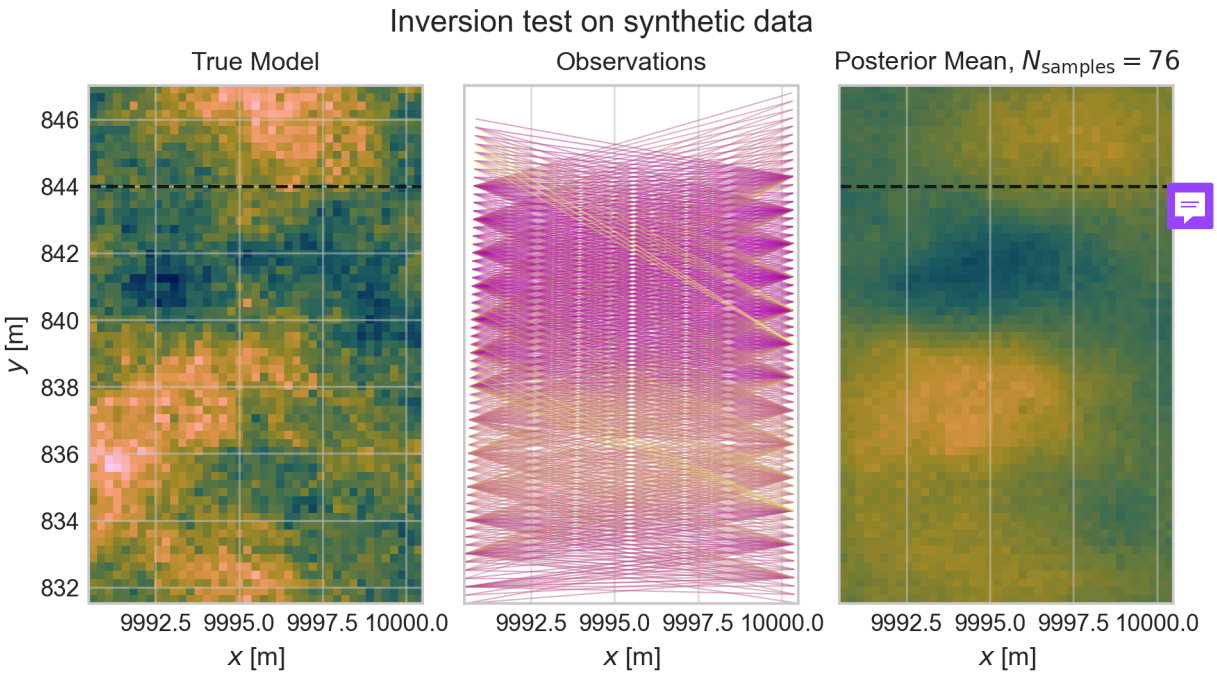


Figure 13: Reconstruction of a synthetic velocity field drawn from the distribution modelled according to the statistics of the posterior samples.

It is clear that the reconstruction is far from perfect, but it does capture the general structure of the synthetic phantom and exhibits artefacts commonly found in tomographic reconstructions. In particular, we observe a shift in the lighter region in the bottom left corner towards the center in the reconstruction. This is likely a consequence of the non-uniqueness of the inverse problem, where many differenet solutions may produce similar observations.

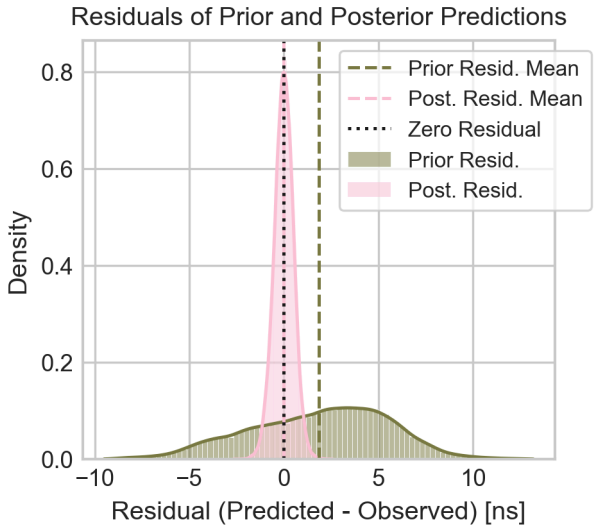


Figure 14: Residuals of the travel time observations for the synthetic experiment.

By computing the residuals for over all model parameters against the prior and posterior distributions we are able to construct the histograms shown in Figure 14, which reveals that the posterior distribution is narrow, approximately normally distributed, and centred around zero, while the residuals of the prior show a biased and much wider distribution.

This gives credence to both the veracity of the eikonal solver and the correctness of the implementation of the sampling algorithms in `nanopinv`, as well as the overall validity of this probablistic approach to solving the inverse problem.

3.5 Porosity and Total Water in Place

Before turning to the results of our method applied to the problem in full, we briefly introduce the following empirical relation with which the porosity, Φ, of the subsurface may be estimated

$$\Phi = \exp(-41.7v + 2.03). \tag{16}$$

where v is the phase velocity field obtained by the inversion. The origin of this relation is unknown to the author and as such is simply assumed sufficiently accurate for the purposes of this report.

We may then introduce the metric of Total Water In Place (TWIP), measures the total area of water contained in the subsurface given a porosity distribution beneath the water table. It is defined simply as the integral of the porosity over the area of interest:

$$\begin{aligned} \mathrm{TWIP} &= \int \Phi(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} \\ &= \sum_{i=1}^{N_x}\sum_{j=1}^{N_y} \Phi_{ij}\Delta x \Delta y \end{aligned} \tag{17}$$

where N_x, N_y refer to the number of discretisation points in the x and y directions respectively, and $\Delta x, \Delta y$ are the corresponding grid spacings.

4 Results

Having now established the theoretical background and validated the `nanopinv` implementation on parts of the dataset, we set up a parallel tempering sampler using a proposal distribution constructed using the prior distribution obtained in Section 3.2.a and Preconditoned Crank-Nicolson algorithm described Algorithm 1. We again rely on the assumption of uncorrelated and normally distributed data errors in order to arrive at the Gaussian likelihood function described in Section 3.1.a.

The sampler is then initialised following the exact procedure described in Section 3.4.d and subsequently iterate through 1 000 steps after tuning to ensure the chains are in equilibrium at the final parameters.

We then record a further 25 000 steps and plot the diagnostic traces in figures 15, 16, and 17. We note that the proposal rate is steady at the target rate of 25% while the swap acceptance rate has the correct mean, but has large fluctuations for the low temperature chains. As a consequence, their auto-correlation remains relatively high compared to the higher temperature chains, though still significantly shorter than those of the regular Extended Metropolis sampler presented in Section 3.4.a.

Based on the auto-correlation plot we opt to thin the accepted samples to a minimum interval of 300 iterations, which produces 81 uncorrelated samples from the posterior distribution.

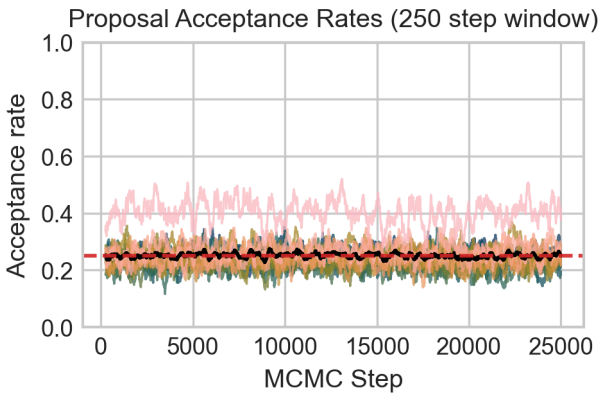


Figure 15: Trace of proposal acceptance rate for samples during sampling. Red dashed line indicates 25% and black line denotes mean acceptance rate across all chains.

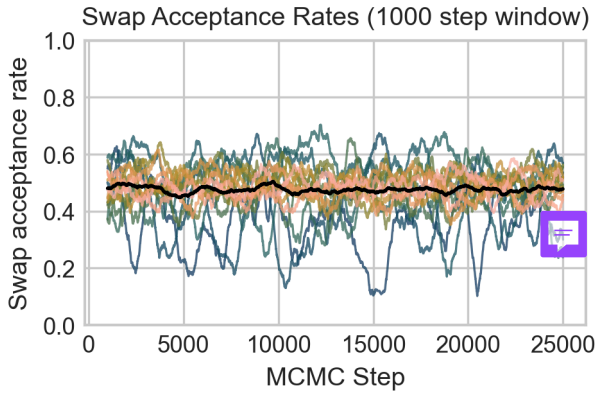


Figure 16: Trace of swap acceptance rate for samples during sampling. Black line denotes mean acceptance rate across all chains.

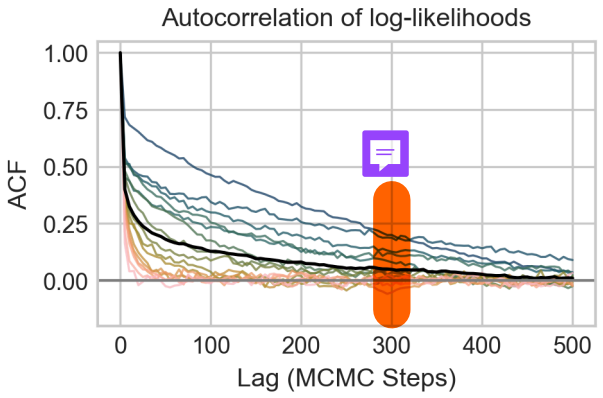


Figure 17: Autocorrelation of the samples during sampling. Black line denotes mean across all chains.

We observe what appears to be good resolution of the velocity field in Figure 18, though the true velocity field is not known and thus it is difficult to make any quantitative statements about the quality of the reconstruction.

When compared against the observations shown by the colored rays in Figure 19 we can see that the reconstructed velocity field is qualitatively consistent with the observed travel times. This is further supported when considering the residuals as shown in Figure 20. The residuals of the posterior distribution are observed to be normally distributed and centered around zero and the width of the distribution is significantly narrowed compared to the prior distribution.

Lastly, we produce estimates of the porosity and total water in place (TWIP) using Equation 16 and Equation 17, whereby we find the porosity field and the TWIP in the water table for each sample in the posterior. By then taking the expectation and standard deviation of the TWIP values, we find the total water in place for the subsurface beneath the water table to be $(27.02 \pm 0.11)\,\mathrm{m}^2$. Recall that the water table was given to be the region below elevation $y_* = 844.0\,\mathrm{m}$,

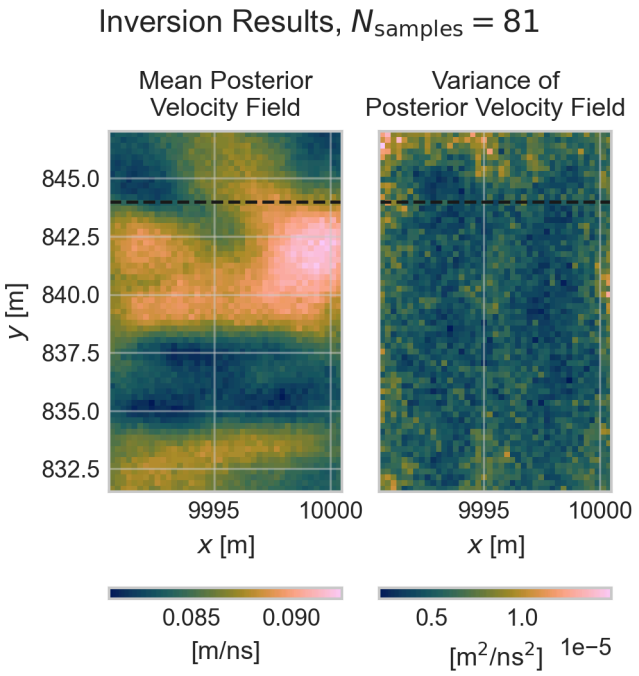


Figure 18: Inversion results for the parallel tempering sampler. Dashed black line denotes water table.

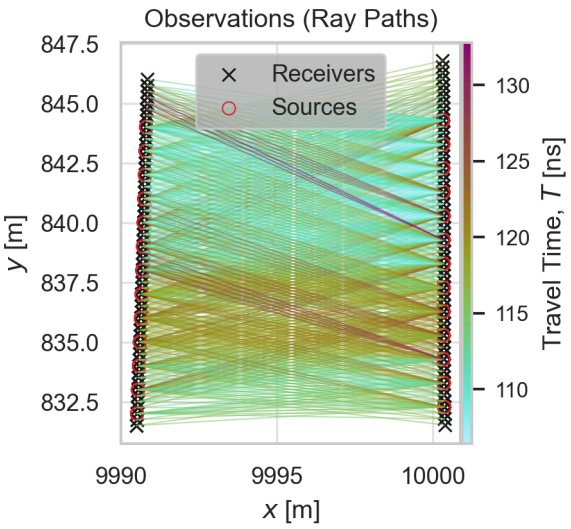


Figure 19: Source and receiver locations with observed travel times overlaid on the model space.

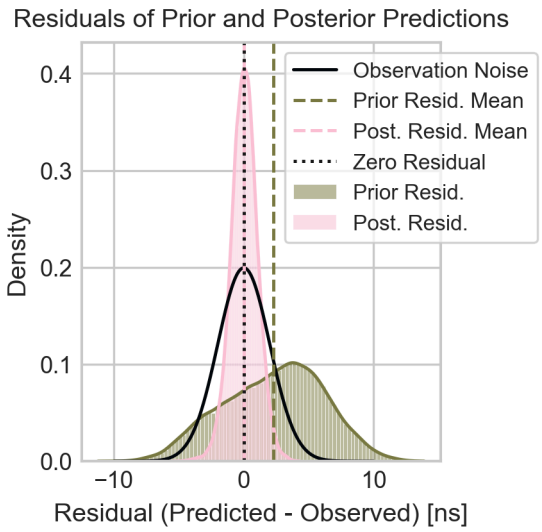


Figure 20: Residuals of the travel time observations for the parallel tempering sampler.

4.1 Effect of solver errors

All results shown so far have been made with the second-order FSM solver introduced in Section 3.3.b, but similar experiments were carried out using the first order FSM solver and both the first and second-order FMM solver from `scikit-fmm` [10]. Remarkably, the 2nd order solvers produce nearly identical results, as may be seen by comparing second-order FMM result shown in Figure 22 with the 2nd order FSM result in Figure 21. Similarly, the first order methods shown in Figure 23 and Figure 24 are in good agreement with each other, but produce significantly different results to the second-order methods.

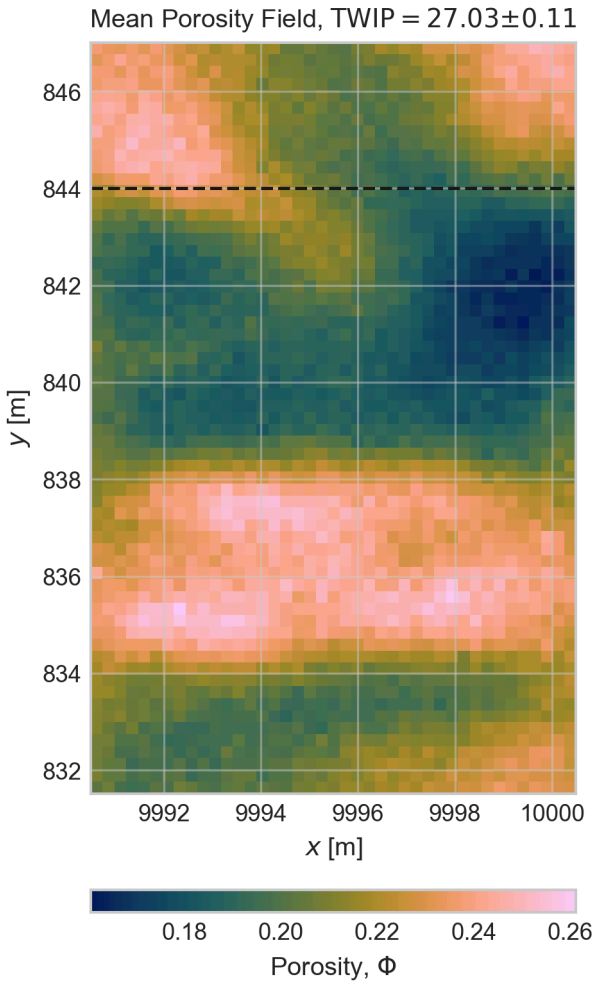


Figure 21: Estimated porosity distribution obtained by applying the empirical relation in Equation 16 to the velocity field obtained from the inversion.

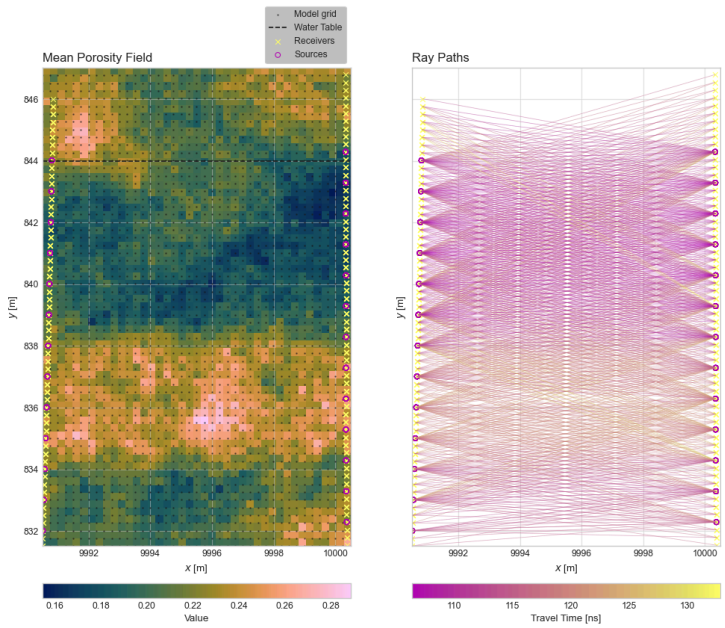


Figure 22: Inversion results obtained using the second-order FMM solver from `scikit-fmm`. Only 15 samples were used and they were averaged to produce a single velocity from which the porosity is computed. This should be improved, but was not due to time constraints.

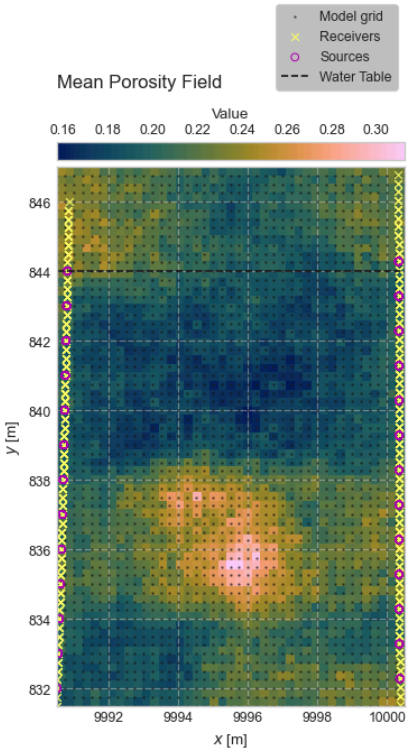


Figure 23: Inversion results obtained using the first order FMM solver from `scikit-fmm`. Similar caveats to Figure 22

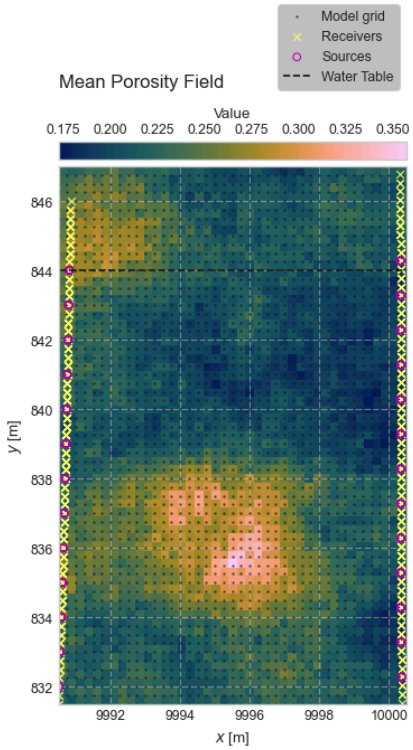


Figure 24: Inversion results obtained using the first order FSM solver. Similar caveats to Figure 22

5 Discussion

5.1 Ill-posedness of the problem

As evidenced by the results of Section 3.4.e covering the synthetic test set, the tomographic problem is highly non-unique and the limited angles of the radar rays negatively affect the resolution in the horizontal direction. Given the apparent smoothness and convexity of the likelihood landscape as seen using the regular Extended Metropolis algorithm in Section 3.4.a, the inversion appears to be relatively stable. As such, the ill-posedness of the inverse problem arises primarily from its lack of uniqueness.

5.2 Overfitting

We observe in Figure 20 that the residual distribution is in fact smaller than the a priori data noise. This is an indication that the model is overfitted to the data set, which may be unsurprising given the problem is significantly under-determined with $41 \times 63 = 2583$ parameters and only $N_d = 758$ data points in the observations. However, the effective parameter space is severely shrunk by the covariance structure of the prior, which acts as a regulariser, thus preventing the many parameters of the model from entering as full degrees of freedom.

5.3 Prior Assumptions

With this in mind, we hypothesise that the over-confidence of the obtained posterior is due to invalid assumptions about the prior distribution. Initial suspicion fell on the non-zero nugget, but enforcing a zero nugget did not change the residual histograms meaningfully.

The modelled posterior appears to violate the assumption of isotropy made for the prior distribution. This assumption also fails on physical grounds, as domain knowledge suggests that we would expect the some horizontal stratification of the velocity field due to the geological processes that formed the subsurface. This causes the chosen covariance structure of the prior to be a poor fit for the true distribution.

The violation of these prior assumptions may be a significant contributor to the overfitting of the model to the data as observed in the residual histograms in Figure 20.

5.4 Solver Errors

Further, the disagreement between the first and second-order solvers observed in Section 4.1 suggests that the truncation error of the solvers is substantial and does introduce significant biases in the reconstruction. This violates the assumption that the data errors dominate the residuals and thus the likelihood obtained in Section 3.1.a does not correctly normalise these residuals. This further contributes to the overfitting of the model to the data.

5.5 Experimental Limitations

As initially highlighted in Section 3.3, it is likely that the high-frequency assumption that leads to the eikonal equation is violated for the GPR data used in this experiment. Without further information about the experimental parameters, it is difficult to gauge whether the data is collected in the high-frequency limit and what the volume of the first Fresnel zone may be. As such, it may be that the spatial structure of the velocity field is substantially finer than the resolution of the data, which would lead to smoothening of the reconstructed velocity field. The obtained velocity fields are relatively smooth and have characteristic length scales at the same order of magnitude as the resolution limit hypothesised in Section 3.3.

Further, the propagation of the radar waves through the subsurface happens in three dimensions, but the model is limited to two dimensions. This implies that the rays may propagate through unmodelled regions of the subsurface which may bias the reconstruction from the data. This further contributes to the ill-posedness of the problem due to a lack of uniqueness.

5.6 Efficiency of Parallel Tempering

The parallel tempering scheme requires substantially more computational time to tune, which may not outweigh its benefits for this particular problem. It allows for efficient sampling of the whole model space, but as observed in Section 3.4.e, the model space appears to be smooth and convex, thus leading to the regular Extended Metropolis algorithm being able to competently explore the model space. These parallel, cold chains do suffer from substantially longer correlation times than what can be achieved with parallel tempering, but have the significant advantage that all chains produce samples that are candidates of the posterior distribution, and additionally are much simpler and faster to tune. For this reason, we find that the regular Extended Metropolis algorithm is a sufficient choice for this problem.

6 Further Work

In addition to further investigating the areas highlighted in the discussion above, there are many outstanding questions and avenues for further work that arise from this project. In particular, the implementation of additional sampling algorithms such as Hamiltonian Monte Carlo (HMC) and the No-U-Turn Sampler (NUTS) would be a useful addition to `nanopinv` and may provide further efficiency improvements over the current parallel tempering scheme.

Additionally, the benefit of the parallel tempering scheme may be demonstrated on a problem with a highly non-convex likelihood landscape on which regular Extended Metropolis would struggle to explore the model space effectively.

Performance comparisons against existing and alternative implementations of probablistic inversion were initially envisioned as part of this project, but were ultimately abandoned due to time constraints.

Similarly, the supremacy of massively-parallel hardware such as GPUs for solving the forward problem and sampling the distribution was observed by the author during the project, but has yet to be rigourously benchmarked in the manner required to be included in this report.

7 Conclusion

We conclude that the `nanopinv` implementation and the employment of both standard extended Metropolis and parallel tempering samplers are successful in producing a reconstruction of the electromagnetic velocity field within the subsurface of the test site at the Boise Hydrogeophysics Research Site (BHRS). Using an a priori empirical relation between velocity and porosity we can further present an estimate of the posterior porosity distribution at the experimental site and produce estimates of the total water in place (TWIP) beneath the water table, which was found to be $(27.02 \pm 0.11)\,\mathrm{m}^2$.

It should be noted that the observed discrepancy between the residuals of the posterior distribution and the a priori data noise is an indication of overfitting, which is believed to be a consequence of violations in the assumptions of the prior distribution and the potential presence of significant truncation errors in the second-order solvers used to compute the forward model. Consequently, the variance of the obtained posterior distribution is believed to be underestimated and thus the confidence in the obtained estimate of the TWIP is likely overestimated.

Lastly, we have demonstrated a performant implementation of probabilistic inversion that can easily leverage massively parallel hardware such as GPUs to solve the forward problem and sample the posterior distribution.

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A Appendix

A.1 Cautious Student Feedback on Assignment

The assignment description is relatively sparse with details about the experiment and the experimental design. It is difficult to come up with this after the fact without knowing where the data came from.

- The assignment incorrectly states that there are 758 sources and receivers. The correct numbers are given in Section 2.
- The GPR carrier frequencies are not given, which makes it difficult to reason about the validity of the high-frequency assumption that arises in the derivation of the forward model through the eikonal equation.
- The dates and location of the data collection are not available.